



Amphiphilic marine coating systems of self-stratified PDMS-PEG surfaces with an epoxy-polyurethane matrix

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Abstract Marine coatings protect submerged surfaces from the negative effects of biofouling. In this work, we demonstrate a new method to prepare self-stratified, amphiphilic glycidyl-carbamate (GC)-based (epoxy urethane-based) coatings (AmpSiGC coatings) that have fouling-release properties making them suitable for marine use. The prepared coating systems are unique and durable in character as the bulk coating takes advantage of both epoxy and urethane functionalities while the surface is comprised of both hydrophilic and hydrophobic domains, granting it an amphiphilic characteristic. The experimental approach aimed to evaluate several factors that influence coating performance, including molecular weight of poly (ethylene glycol) (PEG) and PDMS moieties, ratio of hydrophobic (PDMS) and hydrophilic (PEG) components in the system, and the effect of different curing agents. The results demonstrated that polymeric chains of 10,000 $\overline{M}_n$ PDMS and 750 $\overline{M}_n$ PEG at 10–15 wt.% each offer substantially improved or comparable fouling-release performance in comparison to commercial marine coatings. This paper reports on the facile synthesis and characterization of the GC resin and

GC prepolymers using FTIR and epoxy titrations; surface characterization of the coatings using ATR-FTIR, XPS, and AFM; and fouling-release assessment of the surfaces using laboratory biological assays with the barnacle *Amphibalanus amphitrite*, the algae *Ulva linza* and *Navicula incerta*, and the bacteria *Cellulophaga lytica*. Several of the AmpSiGC coatings exhibited promising performance, which were better or comparable to the internal and commercial reference coatings. The performance of the systems was dependent on all of the factors considered in this study.

Keywords Siloxane-polyurethane, Epoxy coating, Marine coating, Fouling-release, Carbamate coating, Self-stratification, Biofouling

Background

Marine biofouling is caused by the settlement and growth of marine organisms on submerged surfaces.¹ The adverse consequences include increased drag, reduced ship maneuverability, high fuel consumption, and the transportation of nonindigenous species around the world.^{2,3} As an example of the costs involved, the US Navy currently spends \$56M per year to contend only with biofouling on a class of mid-sized naval surface ship (Arleigh Burke-class destroyer DDG-51), and it is projected the hull fouling maintenance cost for this class will be over \$1B for a 15-year period.⁴ Marine biofouling is a complex problem involving more than 4000 marine organisms, having many different settlement strategies and mechanisms of adhesion.^{1,2,5} Ships have been around for thousands of years and humans have developed a variety of systems to combat biofouling. Among the early protective coatings, copper alloys and lead sheaths were applied to ships' hulls, but they had limitations in terms of metallic corrosion and availability of re-

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sources. Biocide-containing antifouling (AF) paints, such as tributyltin-based self-polishing coatings were introduced in the mid-1900s. However, many of these systems caused undesired toxic effects on aquatic environments and despite their effectiveness as antifouling coatings, tin-containing formulations were banned for application on large vessels by the International Maritime Organization (IMO) in early 2000s, motivating the development of nontoxic antifouling and fouling-release (FR) coating systems.^{2,6}

Current FR coatings typically contain copper-based biocides with organic booster biocides to manage biofouling. Copper is less toxic than organotin compounds, but still has potential to poison ecosystems. Alternatively, FR systems offer a nontoxic and environmentally friendly solution to tackle biofouling. Instead of leaching biocides, FR systems function by minimizing the adhesion strength of biofoulants to the surface of a structure and thus facilitating their release under hydrodynamic pressure.^{1,2} Thus, in an ideal world, FR systems would be favored due to the absence of biocides in their design. FR systems are generally made from elastomeric, low surface energy materials such as polydimethylsiloxane (PDMS) and fluoroalkyl polymers.^{2,7} However, these materials often suffer from poor mechanical durability and require a tie coat to attain good adhesion to a surface.^{2,5} Viable alternatives are hydrophobic siloxane-polyurethane (SiPU) FR coatings, which are robust, with good mechanical durability and are able to attach strongly to a wide range of surfaces without the need for a tie coat. The SiPU FR system takes advantage of self-stratification of the nonpolar low-surface-energy PDMS over the polar PU layer, while both layers are connected through covalently bonded crosslinks.^{8,9}

Although hydrophobic FR systems, such as the SiPU coatings, have shown potential against many marine biofouling organisms, they often fail against slimes.⁵ These slimes are frequently dominated by diatoms and bacteria which can adhere strongly to surfaces and cause a significant drag increase on the hull of a ship.¹⁰ In efforts to improve performance of PDMS-based FR coatings, there has been a number of approaches where functionalized nanoparticles of various chemical moieties such as nanomagnetite/silicone, multiwall carbon nanotubes, and graphene oxide were introduced to a coating matrix.^{11–22} Recent advances in FR coatings have included the incorporation of amphiphilic characteristics, which reduce the tenacity of slime adherence. The literature has shown that obtaining the optimum hydrophilic:hydrophobic balance is critical to attain desirable fouling-release coatings.^{2,23–27} As a result, researchers have extensively investigated various methods and chemistries to access reliable amphiphilic surfaces. The approaches are, for example, based on layer-by-layer polyanion-polycation,^{28,29} hyperbranched,^{30,31} UV-cured,³² zwitterionic,^{33–35} self-stratification,^{8,36,37} and polypeptide/peptide-mimic chemistries.³⁸ Recently, PEG-modified amphiphilic SiPU systems have shown desirable FR performance.²⁷ This system is based on an amphiphilic

prepolymer composed of the partially functionalized polyisocyanate resin and acrylic polyol with hydrophobic and hydrophilic moieties. Despite being commonly used in industrial applications, there are many toxicity hazards associated with working with isocyanates. A system is therefore needed that reduces the exposure of workers to isocyanates in the final product while still retaining the benefits of the urethane linkages.

In a previous and closely related study, we reported the desirable performance of such coating systems for antiicing applications.³⁹ Since our previous findings as well as available literature showed that antiicing coatings and marine coatings share similar design strategies,^{27,40,41} we explored the effectiveness of the amphiphilic glycidyl-carbamate (GC)-based coatings as fouling-release coatings in this study. This produced a urethane-based epoxy resin with amphiphilic chains (Fig. 1). GC coatings have been introduced as isocyanate-free systems that retain the desired properties of polyurethanes.^{42–50} Isocyanates are often flagged for their effects on workers' health, however it is challenging to disregard their desired properties that comes from the urethane linkages,^{44,51} thus glycidyl-carbamate (GC) coatings were introduced to deliver both urethane linkage and epoxy chemistry for crosslinking.^{44,45,48} The glycidyl-carbamate functional group consists of an epoxy group positioned adjacent to a urethane (carbamate). GC coatings are composed of a GC resin, which can undergo either curing with an amine⁴⁴ or self-crosslinking.⁴⁸ GC resins are readily synthesized by the reaction of an isocyanate resin with glycidol, generating a urethane/carbamate linkage and introducing epoxy functional groups.

Thermodynamic principles were used to manipulate the self-stratification of low surface energy components, like PDMS-containing materials, and bring them to the surface.⁹ In addition to finding a facile synthesis for attaching PEG and PDMS chains in the same backbone (which usually requires several steps), we investigated three key factors that would pave the way for the effective design of future amphiphilic systems: (1) optimization of the molecular weights of PDMS and PEG for a system; (2) optimization of the total amount of PEG and PDMS in a system; and (3) identification of the effects of crosslinking agents. To this effect, we report on the synthesis and characterization of glycidyl-carbamate (GC)-based resins and amphiphilic prepolymers, and their combination as amphiphilic self-stratified FR coatings that offer desirable fouling-release performance as well as antiicing characteristics.

Experimental

Materials

Isophorone diisocyanate (IPDI) polyisocyanate Desmodur Z4470 BA was supplied by Covestro LLC. Monocarbinol-terminated polydimethylsiloxanes (CT-PDMS) with molecular weights of 5000 $\overline{M}_n$ (MCR-

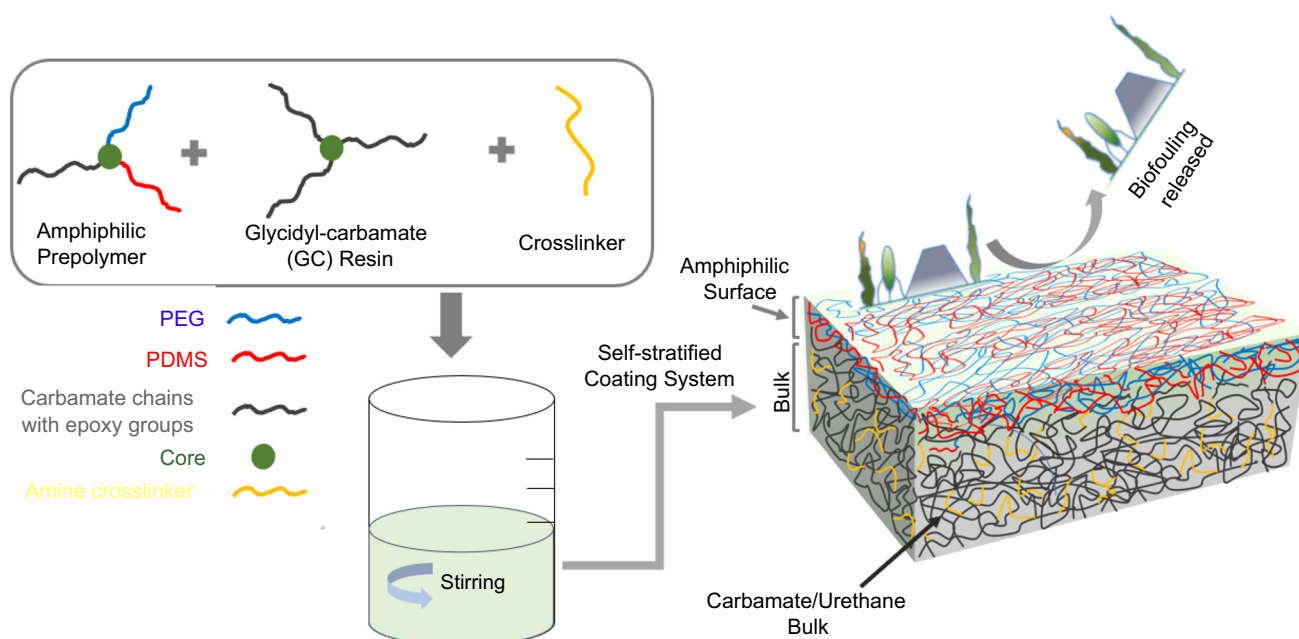


Fig. 1: Schematic illustration of amiphilic self-stratified glycidyl-carbamate coatings for marine fouling-release applications. Figure redrawn from with permission from reference 60. Copyright 2020 American Coatings Association

C18) and 10,000 $\overline{M}_n$ (MCR-C22) were purchased from Gelest, Inc. Glycidol, poly(ethylene glycol) methyl ether (550 $\overline{M}_n$ and 750 $\overline{M}_n$), ethyl-3-ethoxy propionate, methyl ethyl ketone (MEK), methyl amyl ketone (MAK), hydrobromic acid (5.7 M), and dibutyltin diacetate (DBTDAC) were purchased from Sigma-Aldrich. Toluene and glacial acetic acid (for epoxy titration) were purchased from VWR. Ancamide® 702B75 and Amicure® PACM curing agents were provided by Evonik Industries.

AkzoNobel International Paint provided the commercial FR standards Intersleek® 700 (IS-700), Intersleek® 900 (IS-900), and Intersleek® 1100SR (IS 1100). Silicone elastomer, Silastic® T2 (T2) was provided by Dow Corning as another commercial standard. Hydrophobic A4-20 coating (A4-20), a siloxane-polyurethane system, was prepared as an internal control following procedures described elsewhere.⁹ Amiphilic T-10 coating, an isocyanate-based formulation that contained 10 wt.% PEG 750 $\overline{M}_n$ and PDMS 10,000 $\overline{M}_n$, was prepared as described previously.²⁷ Aluminum panels (4" × 8", 0.6 mm thick, type A, alloy 3003 H14), purchased from Q-lab, were sandblasted and primed with Intergard 264 (International Paint) using air-assisted spray application. Multiwell plates were modified using circular disks (1-inch diameter) of primed aluminum.

Design of experiments

This first part of the study examined the effects of four factors: the molecular weights of PEG and PDMS and

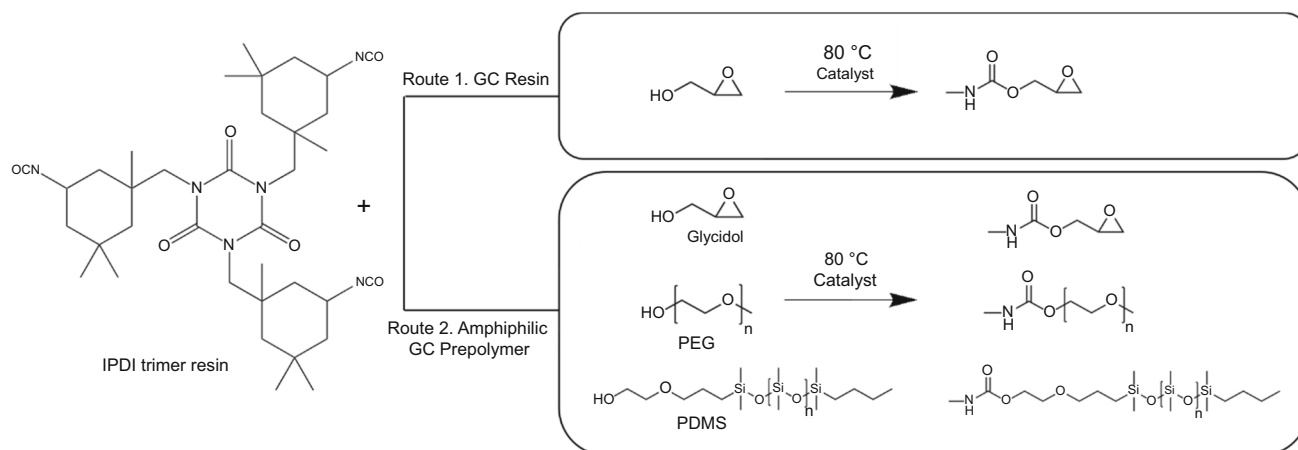
the amounts of each of them used in the coating system. To assess these factors, three overlapping designs were utilized.^{52,53} First, a 2^3 design was followed, involving eight coating formulations. For this design, each of the three factors had two levels: PEG in two molecular weights of 550 $\overline{M}_n$ and 750 $\overline{M}_n$; PDMS in two molecular weights of 5000 $\overline{M}_n$ and 10,000 $\overline{M}_n$; and PDMS and PEG amounts of 5 and 10 wt.% each based on solid contents. These eight formulations are labeled F1–F8 in Table 1. To further evaluate the effect of higher amiphilic amounts in a system, three formulations that contained between 15 and 20 wt.% of PEG 550/750 $\overline{M}_n$ and PDMS 10,000 $\overline{M}_n$ were evaluated. These formulations were varied solely on the amount of PEG and PDMS they contained and are labeled F4, F8, F9, F10, and F11 (Table 1). The second part of the study assessed the effect of crosslinking agent. Accordingly, three additional formulations were introduced, cured with Ancamide®702B75 (F12, F13, and F14). These could be compared directly with their PACM-cured counterparts (F6, F8, and F11) (Table 1).

Synthesis of glycidyl-carbamate resin

Polyisocyanate IPDI trimer Desmodur Z4470 was reacted with glycidol to prepare the glycidyl-carbamate (GC) resin, an epoxidized urethane-linkage-containing resin (Scheme 1—Route 1). The GC resin was produced by reacting the IPDI trimer resin with glycidol in 1:1 ratio of NCO:OH. To synthesize the resin, 13.00 g of IPDI trimer (NCO eq wt. = 253), 2.92 g glycidol, 0.48 g DBTDAC catalyst solution (1% by wt.

Table 1: Coating compositions

Formulation	Formulation type	PDMS MW ($\overline{M}_n$)	PDMS wt. %	PEG MW ($\overline{M}_n$)	PEG wt. %	Curing agent
F1	5-5kPDMS-550PEG	5,000	5	550	5	PACM
F2	5-5kPDMS-750PEG	5,000	5	750	5	
F3	5-10kPDMS-550PEG	10,000	5	550	5	
F4	5-10kPDMS-750PEG	10,000	5	750	5	
F5	10-5kPDMS-550PEG	5,000	10	550	10	
F6	10-5kPDMS-750PEG	5,000	10	750	10	
F7	10-10kPDMS-550PEG	10,000	10	550	10	
F8	10-10kPDMS-750PEG	10,000	10	750	10	
F9	15-5kPDMS-750PEG	5,000	15	750	15	
F10	15-10kPDMS-750PEG	10,000	15	750	15	
F11	20-10kPDMS-750PEG	10,000	20	750	20	
F12	10-5kPDMS-750PEG-RT	5,000	10	750	10	Anacamide 702B75
F13	10-10kPDMS-750PEG-RT	10,000	10	750	10	
F14	13-10kPDMS-750PEG-RT	10,000	13	750	13	

**Scheme 1: Synthesis of glycidyl-carbamate resin (Route 1) and amphiphilic GC prepolymer (Route 2). Reproduced with permission from reference 60. Copyright 2020 American Coatings Association**

in MAK), and 7.95 g toluene were added together in a 50-mL three-neck flask, equipped with a magnetic stirrer, nitrogen inlet, and temperature controller. The reaction was carried out at 80°C for 2 h. A reflux condenser was used.

Synthesis of amphiphilic glycidyl-carbamate prepolymers

GC-functional amphiphilic prepolymers were synthesized by attaching epoxy functional groups and PEG and PDMS chains onto the polyisocyanate IPDI trimer Desmodur Z4470 resin (Scheme 1—Route 2). The ratio of NCO groups to the combined OH groups was 1:1. Glycidol was used to functionalize 33.3% of the NCO groups and the remaining 66.6% isocyanate

groups were converted to urethane linkages by reaction with PEG and PDMS chains. PEG and PDMS were added in equal weight ratios to meet the 66.6% required molar ratio. For example, to synthesize the prepolymer 5kPDMS-550PEG, PEG 550 $\overline{M}_n$ (2.50 g) was first diluted in toluene (2.50 g) in a 50-mL flask. PDMS 10,000 $\overline{M}_n$ (2.50 g) was then added to the flask and mixed robustly with vortexing for 2 min. Glycidol (0.13 g) was added and mixed for another 2 min. Following this, IPDI trimer resin (1.84 g) and DBTDAC catalyst solution (1% by wt. in MAK) (0.35 g) were added to the flask. The reaction was carried out at 80°C for 2 h. The flask was equipped with a magnetic stirrer, nitrogen inlet, and temperature controller. A reflux condenser was used. In theory, the synthesized prepolymer contained 38.5 wt.% PEG and 38.5 wt.% PDMS.

Epoxy equivalent weight titrations

Epoxy equivalent weight (EEW) titrations were used to verify EEW content for both resin and prepolymers. The titration was conducted following ASTM D1652. A sample of resin or prepolymer (1–2 g) was weighed out in an Erlenmeyer flask. The sample was dissolved in 15 mL of chloroform and an indicator solution of 1% crystal violet in glacial acetic acid was added (4–6 drops). The content of the flask was titrated with a standardized solution (0.1 N) of hydrobromic acid in glacial acetic acid until the blue solution displayed a color change to blue-green marking the endpoint of the reaction. A blank titration was also carried out using 15 mL pure chloroform without sample. Three replicates were recorded for each resin/prepolymer. EEW was calculated using the recorded information and the percent solids content.

Percent solids determination

The percent solids were determined using ASTM 2369 so that the volatile content of resins or prepolymer could be measured. In summary, a known amount of resin/prepolymer sample (1–2 g) was weighed out in an aluminum pan and covered with isopropyl alcohol. The pan was placed in an oven at 120°C for 1 h. After removal from the oven, the pan was weighed again, and the percent solids was determined. Three replicates were recorded for each resin/prepolymer.

Fourier transform infrared spectroscopy

Fourier transform infrared (FTIR) spectroscopy was employed to characterize GC resin and prepolymers, using a Thermo Scientific Nicolet 8700 FTIR. The resin or prepolymer was spread as a thin layer on a potassium bromide (KBr) plate to collect the spectrum.

Coating formulation and curing

The GC resin and amphiphilic GC prepolymer were mixed in a vial and stirred for 24 h. Curing agent (i.e., PACM) was then added to the vial and stirred for another 20 min. The formulation was allowed to stand for 15 min to ‘sweat-in’ before application. The formulation was applied to 8" × 4" aluminum substrates using a drawdown bar with a wet film thickness of 80 µm. Coatings with PACM curing agent were oven-cured at 80°C for 45 min and coatings with 702B75 curing agent were cured ambiently for 8 h. Formulations for the resin and prepolymers that were described earlier are provided in Table 1. To formulate coating 5-5kPDMS-550PEG (F1), for example, 2.85 g amphiphilic GC prepolymer (70.26% solid), 21.57 g GC resin (54.68% solid), and 1.6 g PACM curing agent were used. All outlined formulations for this study

were prepared in the same way. The highest amount of PEG and PDMS in a system using PACM as curing agent was 20 wt.% (Formulation F11) and using 702B75 as curing agent was 13 wt.% (Formulation F14). The recommended amounts of PEG and PDMS were determined to be the maximum concentration of these moieties that did not adversely impact the properties of the coating film. The properties were assessed by a series of mechanical tests on the coatings including hardness, impact, and solvent double rubs.

Surface characterization

A Kruss® DSA 100 (drop shape analyzer) was used to determine the surface wettability of the coatings and their surface energy. For each surface, water and diiodomethane contact angles were measured three times. For each replicate, the static contact angle was measured over 6 min to monitor changes due to the potential interaction of the surface with the water droplet as a function of time. Surface energy for each surface was calculated using the Owens–Wendt method.⁵⁴ Also, slip angle, and advancing and receding water contact angles for each surface were measured using a tilting stage where a 25-µL water droplet was monitored on a coating surface that was tilted at 10°/min. The measured angles and surface energies were auto-calculated using Kruss® Advance software.

Attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR) was used to further characterize the surfaces of the coatings. A Bruker Vertex 70 with Harrick's ATRTM accessory using a hemispherical Ge crystal was utilized to collect ATR-FTIR spectra for a coating.

A Thermo ScientificTM K-AlphaTM X-ray photoelectron spectroscopy (XPS) instrument was used to determine the surface elemental composition of the coatings. The apparatus was equipped with monochromatic Al K_α (1486.68 eV) X-ray source and an Ar⁺ ion source (up to 4000 eV). All of the samples were cleaned to remove trace contaminants; a 2 mm × 2 mm area of the sample was sputtered with a large Ar⁺ ion cluster with a power of 4000 eV using the MAGCIS® cluster gun before analysis. Depth profiling of a coating was evaluated with 20 etch cycles. For each etch cycle, the ion beam was set to 1,000 eV Monatomic Mode with low current and 30 s etch time. After each etching cycle, survey spectra in five replicates were collected at low resolution with a constant analyzer pass energy of 200 eV for a total of 20 ms. For each run, photoemission lines for C1s, N1s, O1s, and Si2p were observed. Spectra were collected at an angle normal to the surface (90°) of a 400-µm area. The chamber pressure was maintained below 1.5×10^{-7} Torr and samples were analyzed at ambient temperature. Atomic concentrations were quantified using the instrument's software.

Atomic force microscopy (AFM) was used to examine the surface topography of the coatings. A

Dimension 3100 microscope with nanoscope controller scanned the surface of experimental coatings, collecting images on a sample area of $100\ \mu\text{m} \times 100\ \mu\text{m}$ in tapping mode. Imaging was done in air under ambient conditions, using a silicon probe with a spring constant (0.1–0.6 N/m) and resonant frequency (15–39 kHz). For each surface, two to three replicates were collected to ensure consistency and accuracy of the data.

Biological laboratory assays

Growth and release of the macroalga *Ulva linza*

A set of 24-well plates containing the test coatings was immersed in water for 28 days to allow the leaching of unreacted materials and equilibration with an aqueous environment. The detailed description of the method to evaluate the FR performance of coatings against *U. linza* can be found elsewhere.⁵⁵ Briefly, after leachate collection, all multiwell plates were equilibrated in 0.22 μm filtered artificial seawater for 2 h. To each well, 1 mL spores of *U. linza* suspension (3.3×10^5 spores/mL) in enriched seawater medium was added.⁵⁶ Spores that settled on the disks were grown for 7 days inside an illuminated incubator at 18°C with a 16:8 light: dark cycle (photon flux density $45\ \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$). There was no washing to remove unsettled spores after settlement. After 7 days, the biomass generated was assessed from a single row of wells (6) from each plate. The chlorophyll was extracted by adding 1 mL DMSO to each well and the fluorescence at 360 nm excitation and 670 nm emission was measured. Fluorescence is directly proportional to the biomass present on each coating surface. FR performance was assessed by exposing two rows of wells to different water-jet treatments (water impact pressures of 10 and 16 psi). Assessment was again made using extraction and measurement of chlorophyll content. The removal of *U. linza* at each pressure was compared with the unsprayed wells that were used to determine initial growth.

Bacterial (*Cellulophaga lytica*) biofilm adhesion

Fouling-release properties toward bacteria were evaluated using retention and adhesion assays described previously.^{57–59} Briefly, a suspension consisting of the marine bacterium *Cellulophaga lytica* at 10^7 cells/mL concentration in artificial seawater (ASW) containing 0.5 g/L peptone and 0.1 g/L yeast extract was deposited into 24-well plates (1 mL/well). The plates were then incubated statically at 28°C for 24 h. The ASW growth medium was then removed and the coatings were subjected to water-jet treatments. The first column of each coating on the 24-well plate (3 replicate wells) was not treated and served as the initial amount of bacterial biofilm growth. The second and third columns (3 replicate wells) were subjected to impact

pressure water-jetting at 10 psi and 20 psi, respectively, for 5 s. Following water-jet treatments, the coating surfaces were stained with 0.5 mL of a crystal violet solution (0.3 wt. % in deionized water) for 15 min and then rinsed three times with deionized water. After 1 h of drying at ambient laboratory conditions, the crystal violet dye was extracted from the coating surfaces by adding 0.5 mL of 33% acetic acid solution for 15 min. The resulting eluates were transferred to a 96-well plate (0.15 mL/coating replicate) and subjected to absorbance measurements at 600 nm wavelength using a multiwell plate spectrophotometer. The absorbance values were considered to be directly proportional to the amount of bacterial biofilm present on coating surfaces before and after water-jetting treatments.³³

Growth and release of the diatom *Navicula incerta*

Laboratory biological assays using the diatom *Navicula incerta* were conducted following a procedure that has been described previously.^{1,55,60} Briefly, a suspension with 4×10^5 cells/mL of *N. incerta* (adjusted to 0.03 OD at absorbance 660 nm) in Guillard's F/2 medium was deposited into each well of a 24-well plate (1 mL per well) and cell attachment was stimulated by static incubation for 2 h under ambient conditions in the dark. Coating surfaces were then subjected to water-jet treatments.⁵⁸ The first column of wells (3 wells) was not water-jetted so that initial cell attachment could be determined and the next two columns of wells (3 wells) were impact water-jetted at 10 psi and 20 psi, respectively, for 10 s. Microalgae biomass was quantified by extracting chlorophyll in 0.5 mL of DMSO and measuring fluorescence of the transferred extracts at an excitation wavelength of 360 nm and emission wavelength at 670 nm. The relative fluorescence (RFU) measured from the extracts was considered to be directly proportional to the biomass remaining on the coating surfaces after water-jetting.

ADULT BARNACLE (*AMPHIBALANUS AMPHITRITE*) ADHESION: An adult barnacle reattachment and adhesion assay was used to evaluate the FR performance of the coatings toward macrofoulers.^{61,62} Coatings prepared on $4'' \times 8''$ panels, after water aging, were utilized for this laboratory assay. Barnacles were dislodged from silicone substrates sent from Duke University (USA) and immobilized on experimental coatings (6 barnacles per coating) using a custom-designed immobilization template. The immobilized barnacles were allowed to reattach and grow for 2 weeks while immersed in an ASW aquarium tank system with daily feedings of brine shrimp, *Artemia* sp., nauplii (Florida Aqua Farms). After the 2-week attachment period, the number of nonattached barnacles was recorded, and the attached barnacles were pushed off (in shear) using a handheld force gauge mounted onto a

semiautomated stage. Once the barnacles were dislodged, their basal plate areas were determined from scanned images using Sigma Scan Pro 5.0 software program. Barnacle adhesion strength (MPa) was calculated by taking the ratio of peak force of removal to the basal plate area for each reattached barnacle. The average barnacle adhesion strength for each coating was reported as a function of the number of barnacles released with a measurable force and that exhibited no visible damage to the basis or shell plates.

Statistical analysis

Analysis of variance for the 2^3 factorial design (Coating F1-F8 included in this design) and completely randomized design (CRD) (all experimental, control, and reference coatings included in this design) were performed in SAS software, version 9.4. The generalized linear model (GLM) procedure with Tukey's method was utilized to compare means of each treatment group under the CRD design. The assessed responses for the analyses were the ease of release of *U. linza* and adhesion strength of the reattached barnacles.

Results and discussion

Amphiphilic coatings contain both hydrophobic and hydrophilic domains on their surfaces and have shown potential for antifouling (AF) and fouling-release (FR) use in the marine environment. Following an earlier study that reported on the preparation of amphiphilic FR coatings based on isocyanate chemistry,²⁷ we developed a system that utilizes a new nonisocyanate crosslinking chemistry while still benefiting from urethane linkages. To this effect, this study explored the development of a FR system based on epoxy-amine crosslinking that is accessible through glycidyl-carbamate chemistry, that is, via conversion of the isocyanate groups of the IPDI trimer resin to epoxy functional groups using glycidol.

A coating formulation was prepared composed of three parts, glycidyl-carbamate (GC) resin (the IPDI-based epoxy resin), amphiphilic prepolymers, and curing agent. The amphiphilic prepolymer based on an IPDI trimer resin was furnished with chains of polydimethylsiloxane (PDMS) and poly(ethylene glycol) (PEG), accounting for 66.6% of the isocyanate groups, while having a partial functionality of 33.3% via its epoxy groups. In order to determine the optimum design for the amphiphilic coatings, a number of factors were compared in each formulation. Molecular weights of PDMS (5,000 $\overline{M}_n$ and 10,000 $\overline{M}_n$) and PEG (550 $\overline{M}_n$ and 750 $\overline{M}_n$) were assessed at two levels for each component to determine how the molecular weight of the amphiphilic chains impact performance. The amounts of PDMS and PEG in the system were

investigated at 5 wt.%, 10 wt.%, 15 wt.%, and 20 wt.% for the effect of amphiphilic concentration on performance. The effect of amine curing agents PACM and 702B75 on performance was also explored.

The complete conversion of the isocyanate-functional IPDI resin to epoxy functional GC (urethane) resin was confirmed with FTIR and epoxy titrations. The FTIR spectrum of the GC resin showed the absence of the signature isocyanate peak at 2250 cm^{-1} and the appearance of the secondary amine (due to formation of the urethane linkage) and ether peaks at 3350 cm^{-1} and 1128 cm^{-1} , respectively (Fig. S1—Red spectrum, see Supplementary Material). The FTIR also showed overlapping stretching for PDMS at 1030 cm^{-1} and PEG at 1105 cm^{-1} , confirming attachment of the amphiphilic chains (Fig. S1—Blue spectrum). Furthermore, epoxy titrations validated the presence of the epoxy functional groups on both the resin and the amphiphilic prepolymers. The epoxy equivalent weight (EEW) values for the samples correlated with their type (Table 2). As expected, the GC resin had a lower EEW value than the amphiphilic-modified GC prepolymer, suggesting the availability of more epoxy functional groups in short segments on the modified isocyanate resin. The value of EEW for the amphiphilic prepolymer was high, supporting the limited presence of epoxy functional groups due to the attachment of PEG and PDMS chains on many of the reactive sites.

The surfaces of the prepared coatings were characterized with a series of experiments. ATR-FTIR was used to qualitatively assess the surface composition of the coatings. The spectra of all the coatings showed similar results and confirmed the amphiphilic nature of the AmpSiGC systems. The comparison of formulations with varying amphiphilicity of 5–20 wt.% PEG/PDMS displayed the presence of siloxane (Si-O-Si) and ether (C-O-C) peaks at $\sim 1030 \text{ cm}^{-1}$ (Fig. 2a—highlighted green) and $\sim 1105 \text{ cm}^{-1}$ (Fig. 2a—highlighted red), respectively. The intensity of the PEG signal relative to the PDMS signal increased as the amphiphilicity content increased, which may be due to the availability of more PEG chains in a system that increases the probability for this moiety diffusing to the surface. Also, a broadened overlapped stretching due to hydroxyl groups (formed due to crosslinking of epoxy and amine groups) and secondary amine (due to urethane linkage) was present at around 3350 cm^{-1} . Furthermore, the FTIR data indicated amphiphilic PEG and PDMS moieties were present on the surface regardless of the curing agent (compare spectra for coatings F11 and F14 crosslinked with PACM and 702B75, respectively) (Fig. 2b).

Contact angle measurements were used to further evaluate the surfaces of the coatings. Figure 3a shows the static contact angles and surface energies over time for the experimental coatings and control coatings (A-4 and T-10). Overall, the coatings displayed a dynamic surface, as the contact angles (both water and diiodomethane) decreased as a function of time (the

Table 2: Epoxy equivalent weight for GC resin and some prepolymers

Material	Theoretical % epoxy on structure (%)	Theoretical % PEG and PDMS on structure (%)	Average EEW $\pm \sigma$
GC resin	100.0	0.0	320.16 $\pm$ 25.20
Prepolymer 10kPDMS-550PEG	33.3	66.6	1987.23 $\pm$ 135.20
Prepolymer 5kPDMS-750PEG	33.3	66.6	1695.20 $\pm$ 95.35

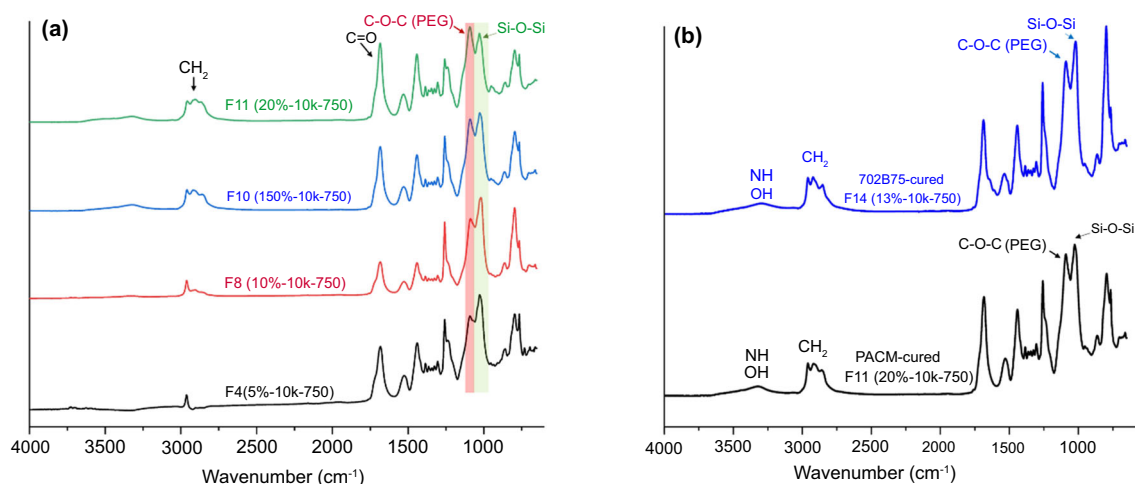


Fig. 2: ATR-FTIR of the AmpSiGC coating surfaces: (a) Spectra of coatings with ranging degree of amphiphilicity from 5 to 20 wt.% of 750 $\bar{M}_n$ PEG and 10,000 $\bar{M}_n$ PDMS; and (b) spectra of F11 and F14 coatings crosslinked with PACM and 702B75 curing agents, respectively. Each spectrum is labeled to reflect a coating ID and its composition. Part (a) is reproduced with permission from reference 60. Copyright 2020 American Coatings Association

values plateaued in 6 min). The decrease in contact angles measured using water (WCA) was more noticeable than for methylene iodide (MICA). The decrease in contact angles was attributed to the presence of hydrophilic domains on the surfaces that swell when interacting with a water droplet, facilitating its spread. This phenomenon was not present for the hydrophobic A4-20 control coating, which did not contain any PEG. Formulations with 10 wt.% PEG and PDMS showed greater dynamic changes than formulations with 5 wt.% amphiphilic components. Also, the higher amount of PDMS in a system caused a higher initial contact angle. This can be seen in the upward trend for formulation F4 with 5 wt.% PDMS, through to formulations F8, F10, and F11 with 10, 15, and 20 wt. % PDMS, respectively. PACM-based coatings, such as F8 and F11, showed lower contact angles than 702B75-based coatings, such as F13 and F14 (highest amount of PEG and PDMS for F14 was 13 wt.% compared to 20 wt.% for F11). The dynamic nature of the contact angles has a direct correlation with the surface energy of the surfaces as illustrated in Fig. 3b. The surface energy values were between 14 and 30 mN/m, depending on the amount of PEG and PDMS in a system and

the type of curing agent. It is noticeable that the surface energy of the amphiphilic systems increases as a function of time, while we did not observe such a behavior for the unmodified PDMS-based A4 control system, demonstrating the amphiphilic surfaces have a dynamic nature that potentially activates in the presence of a solvent such as water. Overall, the greater the changes in contact angle values over time, the higher the change in surface energy. The similarities that were mentioned for contact angle values can also be seen for the surface energy. Coatings with 5 wt.% and 10 wt.% PEG and PDMS (cured with PACM) exhibited properties similar to the T-10 control, while this was not the case for coatings with higher amounts of PEG and PDMS, or with the 702B75 curing agent.

The surfaces of the coatings were further studied by dynamic contact angle measurements to determine advancing/receding contact angles and slip tilting degree (water droplet roll-off angle). Generally, the roll-off angle for the water droplet was between 4° and 8° (Fig. 4a); a value similar to the amphiphilic T-10 coating and considerably different from the hydrophobic A4 control system (with slip angle of 11°). While roll-off angle is an empirical value depending on

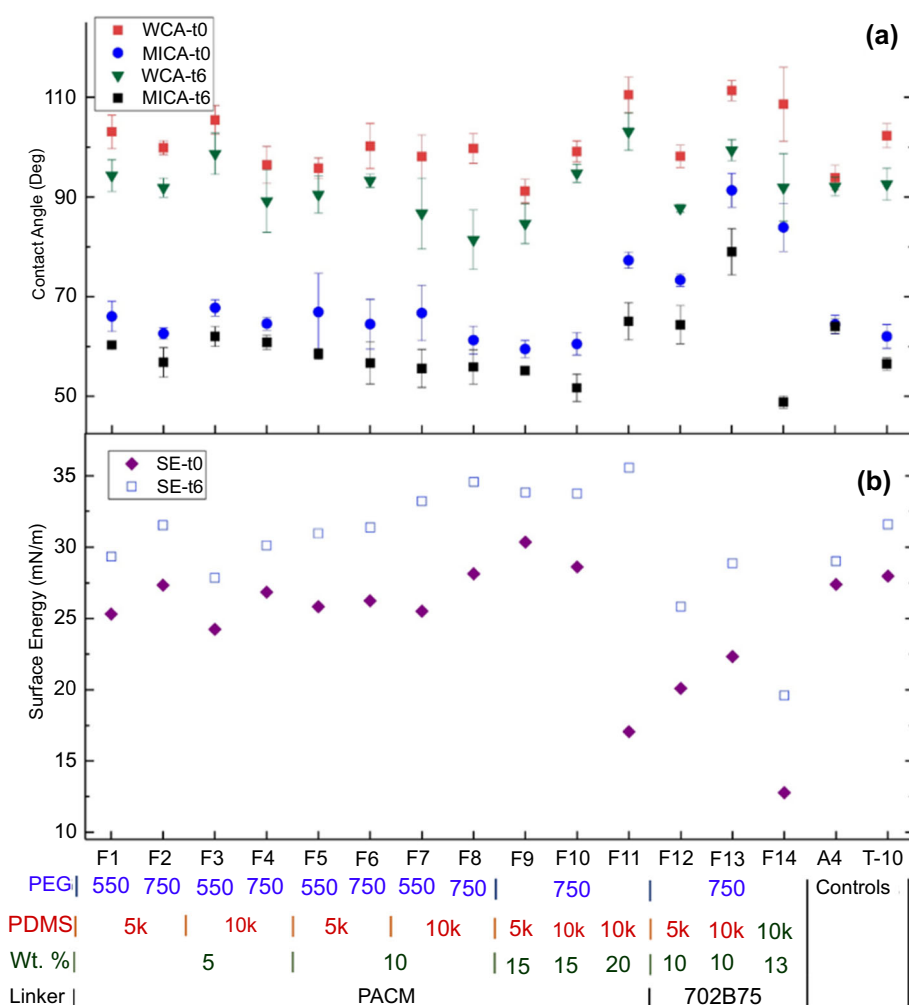


Fig. 3: Static contact angle data for AmpSiGC coatings: (a) Water contact angles (WCA) and methylene iodide contact angles (MICA) as a function of time at 0 min and 6 min, (b) surface energy (SE) of the coatings at 0 min and 6 min, calculated by Owens–Wendt method utilizing the average WCAs and MICAs for each coating. The x-axis is labeled to specify the formulations and its components, including PEG MW, PDMS MW, wt.% of PEG and PDMS, and curing agent type. The error bars in (a) are standard deviations of three measurements

measuring conditions (i.e., droplet size and titling speed), its value conveys how chemical moiety may interact with the surface based on its smoothness. The type of curing agent used also affected the advancing and receding contact angles for a surface (Fig. 4b). Coatings made with the PACM curing agent showed hysteresis values lower than 10° that were comparable to the T-10 and A4 control systems, indicating the presence of a smooth surface. Alternatively, coatings with 702B75 curing agent, and particularly coating F-12, showed a higher degree of hysteresis, indicating the surface is not as smooth as the PACM-containing coatings which may be attributed to differences in rearrangements of surface chemistry.

The surfaces of the amphiphilic systems were characterized by XPS to determine the elemental composition of materials on the surface and in the outer layers of a coating using argon ion milling. The XPS analysis indicated that all of the coatings were

self-stratified, the surface being primarily composed of PDMS-containing materials, while the bulk was constructed of the glycidyl-carbamate polyurethane matrix. As an example, the XPS depth profiling data for coating F8 (Fig. 5a) showed that the surface was dominated by silicon and oxygen atoms. The silicon signal started to decrease beyond a depth of ~ 5 nm, while the concentration of carbon and nitrogen atoms increased due to the crosslinked GC network in the bulk. The XPS depth profiling analysis also suggested that the concentration of amphiphilic domains was directly related to the amount of amphiphilic moieties incorporated in the system (Fig. 5b). This data showed that systems with 15 wt.% and 20 wt.% PDMS had a high concentration of silicon atoms on the surface that extended well into the bulk of the coating. Comparatively, coatings with lower amounts of amphiphilic chains had a lower initial concentration of PDMS on the surface and thus the signal for silicon atoms

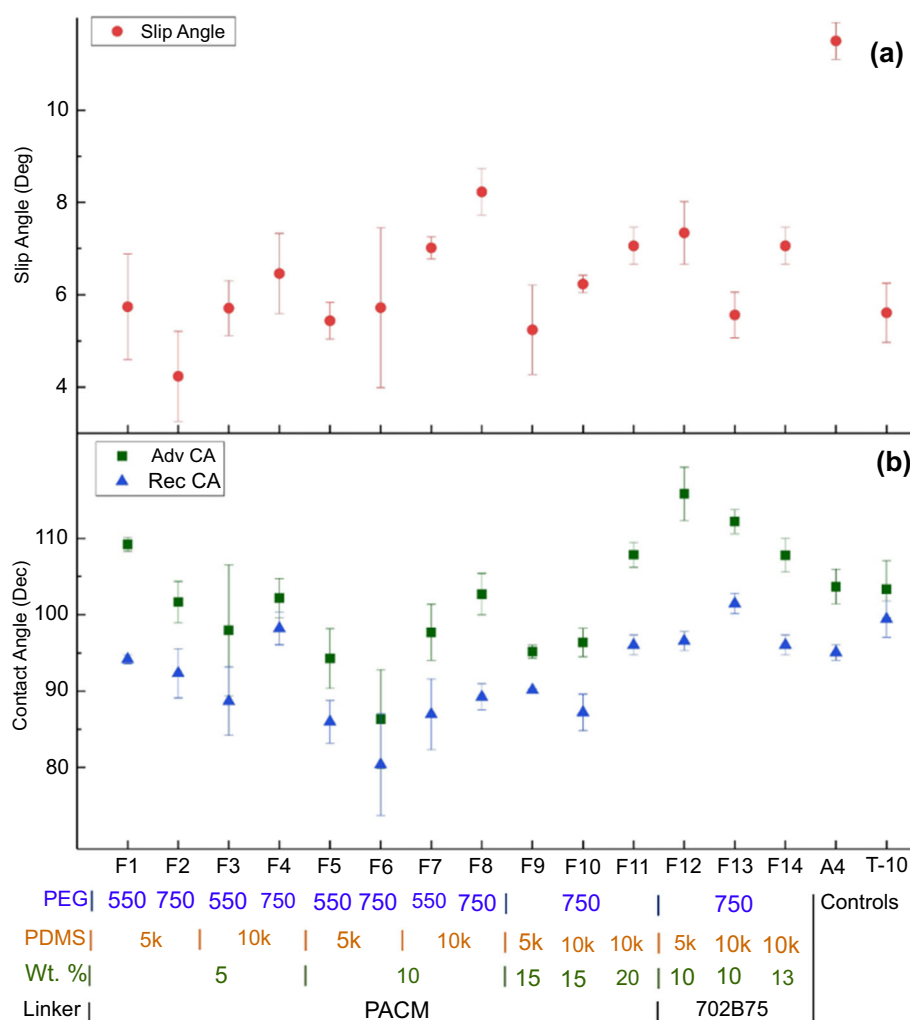
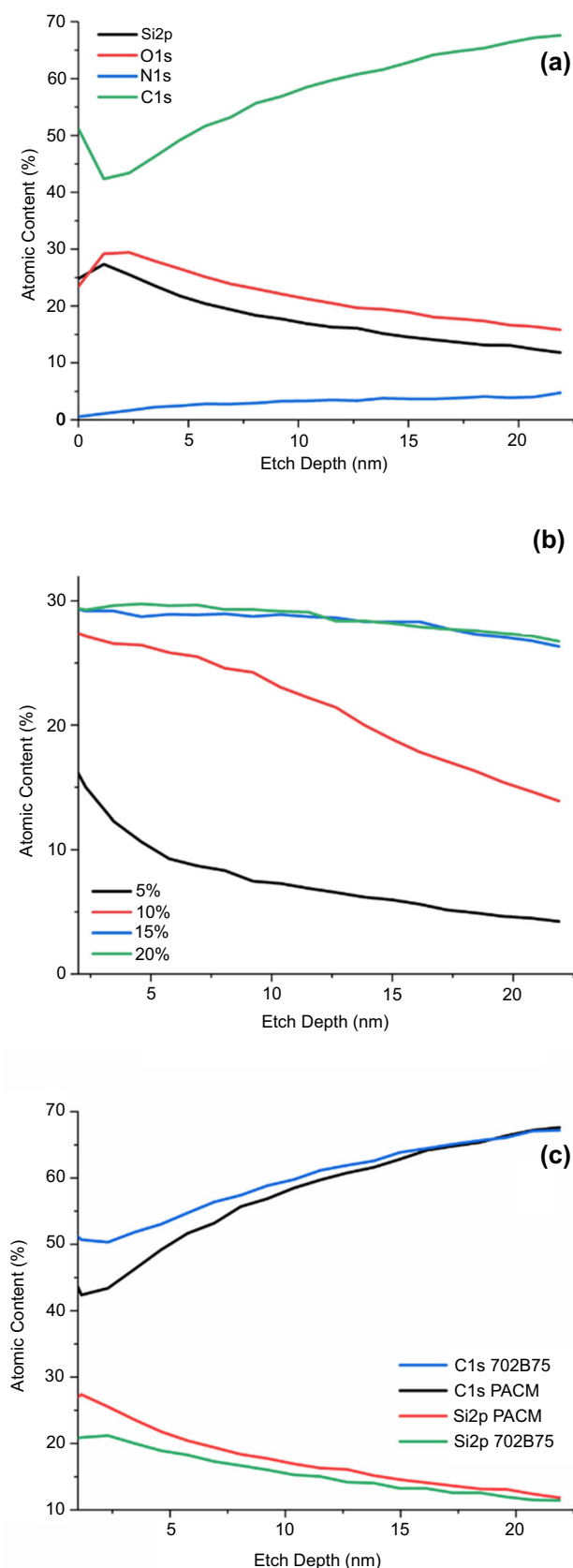


Fig. 4: Dynamic contact angle and slip angle data for AmpSiGC coatings: (a) Water droplet slip angle (roll-off angle for water to slide from a surface), (b) Advancing contact angle (Adv CA) and receding contact angle (Rec CA) data, measured by tilting method. The X-axis is labeled to specify the formulations and its components including PEG MW, PDMS MW, wt.% of PEG and PDMS, and curing agent type. The error bars are standard deviations of three measurements

decreased considerably as a function of thickness. As expected, coatings with 10 wt.% amphiphilic moieties had a higher initial concentration of PDMS on the surface than systems with 5 wt.%. Furthermore, XPS data illustrated that self-stratification occurred for all systems regardless of the type of the curing agent (Fig. 5c), although systems with PACM curing agent had a slightly higher concentration of PDMS at the surface.

AFM was utilized as another technique to characterize the morphology of the surfaces. Generally, soft materials like PDMS appear lighter (having high phase angles) in AFM phase images and harder materials like PEG (harder than PDMS) appear darker (having lower phase angles). The AmpSiGC coatings possessed heterogeneous surfaces composed of light and dark patterns on the surface, signaling the presence of both PDMS (hydrophobic) and PEG (hydrophilic) and therefore an amphiphilic morphology (Fig. 6). The

AFM phase images (Fig. 6) correlate with AFM height images (Fig. S2). Comparing these data with the phase AFM image of the control A4 system (Fig. S3), it is clear that the hydrophobic A4 system lacks the patterns which further support the amphiphilic nature of the AmpSiGC coatings. Coatings with 5 wt.% amphiphilic prepolymers showed smaller microdomains than systems with higher amounts of amphiphilic moieties. Among the 5 wt.% coatings (cured with PACM), it appeared that those containing 5,000 $\overline{M}_n$ PDMS show a well-distributed pattern as small circular domains, while coatings with 10,000 $\overline{M}_n$ PDMS showed stretched-lines domains that were mixed with circular domains. For systems with 10 wt.% amphiphilic moieties (cured with PACM), several changes were noticed including a higher number of circular domains and a wider range of domain sizes. However, the number and size of the heterogenous domains remained relatively similar with 15 wt.% of



◀Fig. 5: XPS data: (a) XPS depth profile analysis for coating F8 (10-10kPDMS-750PEG system), indicating self-stratification of the PDMS-based materials into the surface; (b) XPS depth profile data for silicon atom for coatings F4 (5 wt.%), F8 (10 wt.%), F10 (15 wt.%), and F11 (20 wt.%) that demonstrates effect of amount of amphiphilic prepolymers in a system on the surface composition; (c) XPS depth profile data for coatings F8 (PACM-cured) and F13 (702B75-cured), comparing effect of curing agent type on self-stratification and surface composition. Parts (a) and (b) are reproduced with permission from reference 60. Copyright 2020 American Coatings Association

amphiphilicity. The 702B75-cured coatings such as F12 and F13 displayed a different morphology to the PACM-cured coatings, suggesting the type of curing agent impacts the pattern of a surface. AFM images for coating systems F11 and F14 could not be captured due to limitations with the surfaces, such as oiliness of the interface.

All coatings were immersed in water for 28 days to ensure coatings were stable and any toxic ingredients were leached out before the biological assays to assess FR performance were conducted. Leachate toxicity tests using *C. lytica* and *N. incerta* were carried out as described elsewhere.⁶³ None of the coatings were toxic when compared to positive and negative growth controls (data not reported; available upon request), allowing for further biological evaluations. The fouling-release performance of coating surfaces was evaluated using representative marine organisms including algae *Ulva linza*, bacteria *Cellulophaga lytica*, diatom *Navicula incerta*, and barnacle *Amphibalanus amphitrite*. The performance of coatings for the first marine organisms was assessed on the basis of the amount of biomass or cell numbers remaining on the coatings after exposure to the water jet.

U. linza is a biofouling macroalgae that generally expresses a lower affinity for settlement on hydrophilic surfaces than for hydrophobic surfaces. However, it adheres more tenaciously to hydrophilic surfaces.^{64–66} Consequently, an amphiphilic surface may be a desirable attribute to incorporate in the design of coating systems to reduce biofouling of organisms with this pattern of behavior. The engineered AmpSiGC coatings and all the control and reference coatings showed a relatively similar amount of biomass production by *U. linza* (Fig. 7—Red bars; the higher the bar level, the higher amount of biofouling). Systems cured with 702B75 and systems with high amounts of amphiphilicity (i.e., 15 wt.% and upward) had slightly lower initial biomass levels than the other formulations. The release of *U. linza* was assessed using two water pressures (10 and 16 psi) and the levels of biomass remaining after exposure to the water jets are shown in Fig. 7 and Fig. S4, respectively. At 10 psi, all the systems with 10 wt.% amphiphilic portion or higher showed desirable FR performance and were significantly better than the A4 control system (comparison P-values < 0.05,

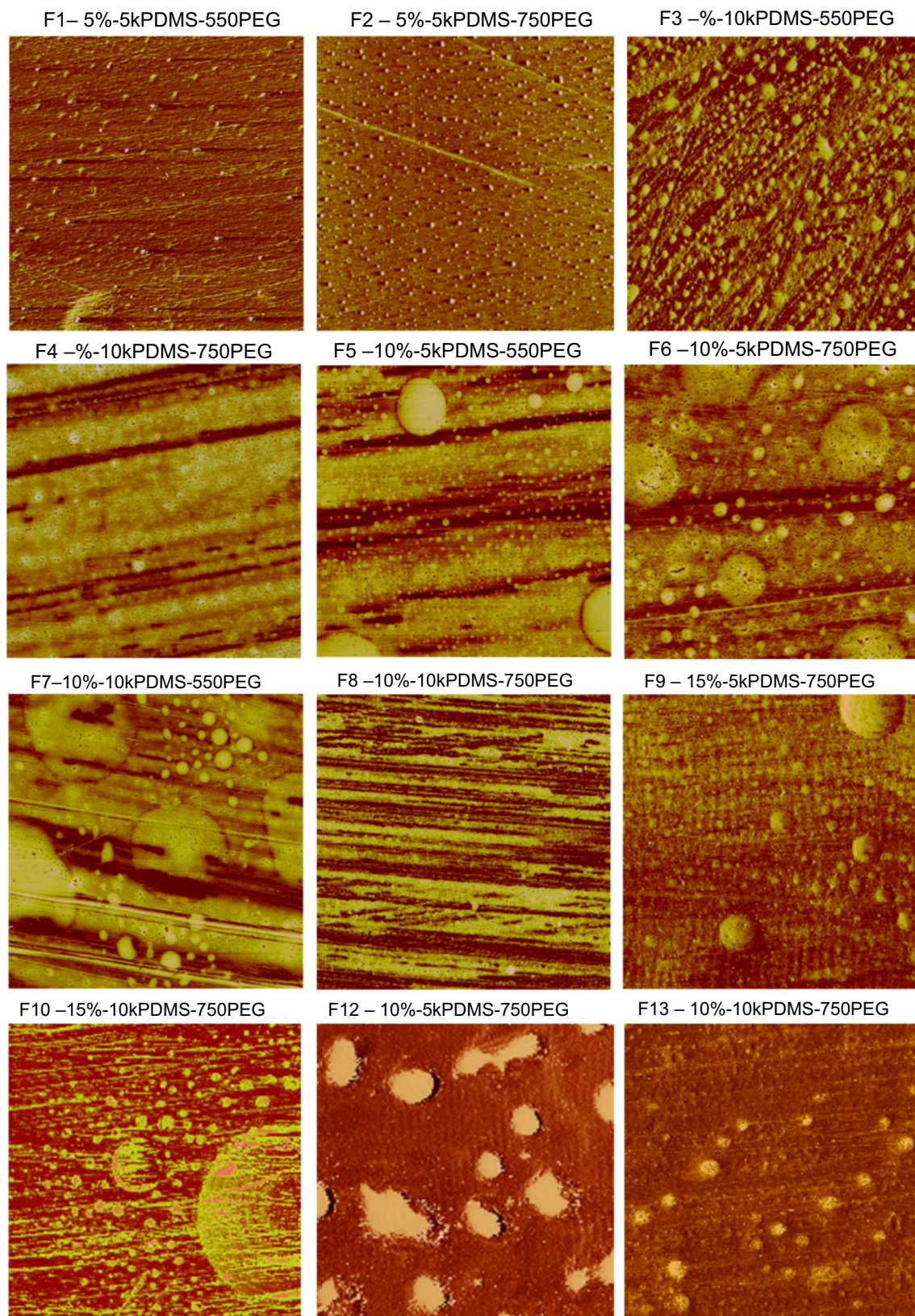


Fig. 6: AFM phase images of AmpSiGC coatings. Each image represents an area of 100 μm x 100 μm . Each label reflects the coating number and composition of an image. Images (F1-F8, F10) reproduced with permission from reference 60. Copyright 2020 American Coatings Association

Tukey's method). Several AmpSiGC formulations, such as F7, F8, F13, and F14, showed better performance than both the IS 1100SR and internal T-10 coatings (comparison P-values < 0.05, Tukey's method) – each of these four coatings was composed of ~10 wt.% PEG (750 $\overline{M}_n$) and PDMS (10,000 $\overline{M}_n$) (Fig. 7). At 16 psi, the FR behavior of the coatings followed a similar trend, except that the extent of the *U. linza* release was improved due to the higher water pressure. The top-performing formulations at 10 psi (F7, F8, F13, and F14) also exhibited better performance at 16 psi than A4 and T-10 systems (comparison P-values < 0.05, Tukey's method), while the release from these was slightly less than from the IS 1100SR (Fig. S4). Overall, the *U. linza* data suggested that AmpSiGC systems are effective FR surfaces.

C. lytica is a known biofouling microorganism with an affinity for a wide range of surfaces, both hydrophilic and hydrophobic.² Biofouling was highest on coatings with 5 wt.% amphiphilicity and decreased on coatings with 10 wt.% or higher amphiphilic content. Specifically, coatings F9-F11 that contained more than 15 wt.% PEG and PDMS and 702B75-cured coatings (F12 and F13 systems) demonstrated the least amount of initial biofouling, which was comparable to the best performing control and reference coatings, suggesting that the amount of PEG and PDMS and the type of curing agent affect the attractiveness of a surface to *C. lytica*. The FR performance of the *C. lytica* biofilm was evaluated at water pressures of 10 psi (Fig. 8) and 20 psi (Fig. S5), and the results for both pressures followed a similar trend. At 10 psi, several coating systems such as formulations F9-F13 outperformed the

internal and commercial controls (comparison P-values < 0.05, Tukey's method). All these coatings contained 15 wt.% or higher PEG and PDMS for PACM-cured systems (F9, F10, and F11) or 10 wt.% PEG and PDMS for 702B75-cured systems (F12 and F13). In comparison, coatings with 10 wt.% amphiphilic chains (F5-F8) cured with PACM did not perform as well as 10 wt.% coatings cured with 702B75, indicating that the choice of curing agent can be instrumental in improving FR performance at low concentrations of amphiphilic moieties. We attribute the better performance of the polyamide-based 702B75 curing agent compared to the PACM curing agent to its more hydrophobic and flexible nature. Additionally, coatings F9-F13 outperformed almost all the commercial marine coatings, including IS-700, IS-900, and IS 1100-SR (Fig. 8; comparison P-values < 0.05, Tukey's method). At 20 psi, the top-performing coatings F9-F13 demonstrated similar trends in removal with higher fouling-release at the higher water pressure (Fig. S5; comparison P-values < 0.05, Tukey's method). Overall, the data for *C. lytica* suggest that the AmpSiGC surfaces deliver a desirable FR performance, which is better than that of the previously established hydrophobic A4 and amphiphilic T-10 systems.

The slime-forming diatom *N. incerta* is another major biofouling organism that typically attaches to hydrophobic surfaces.^{64,65} The extent of diatom biofouling for systems that contained between 5 and 10 wt.% amphiphilic chains and which were cured with PACM was greater than on coatings with higher loadings of PEG and PDMS, which were 702B75-cured, or any of the controls and reference coatings

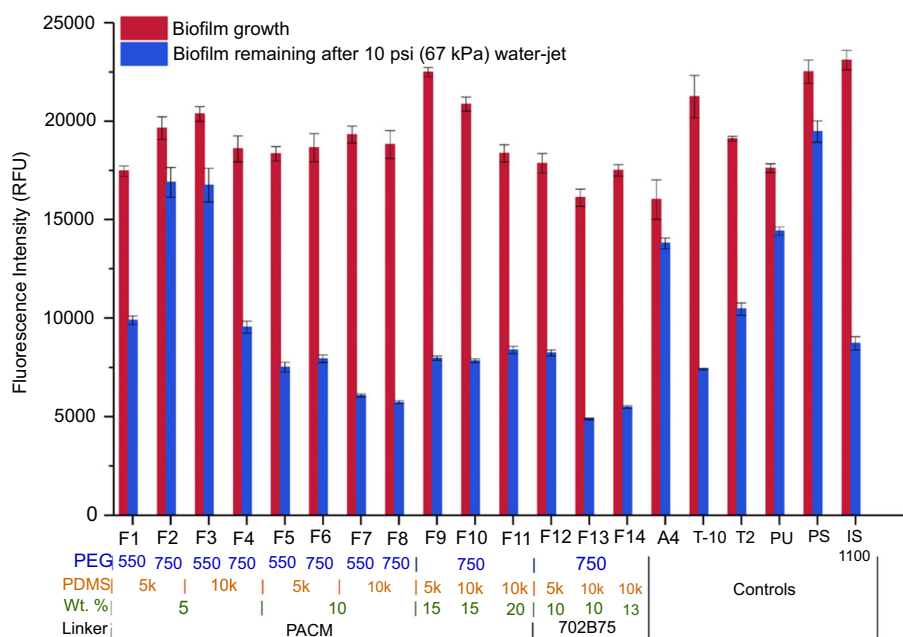


Fig. 7: *U. linza* data for biofilm growth (red bar) and release at 10 psi (blue bar). The x-axis is labeled to specify the formulation and its components; PEG MW, PDMS MW, wt.% of PEG and PDMS, and curing agent type. The error bars are standard deviations of six measurements

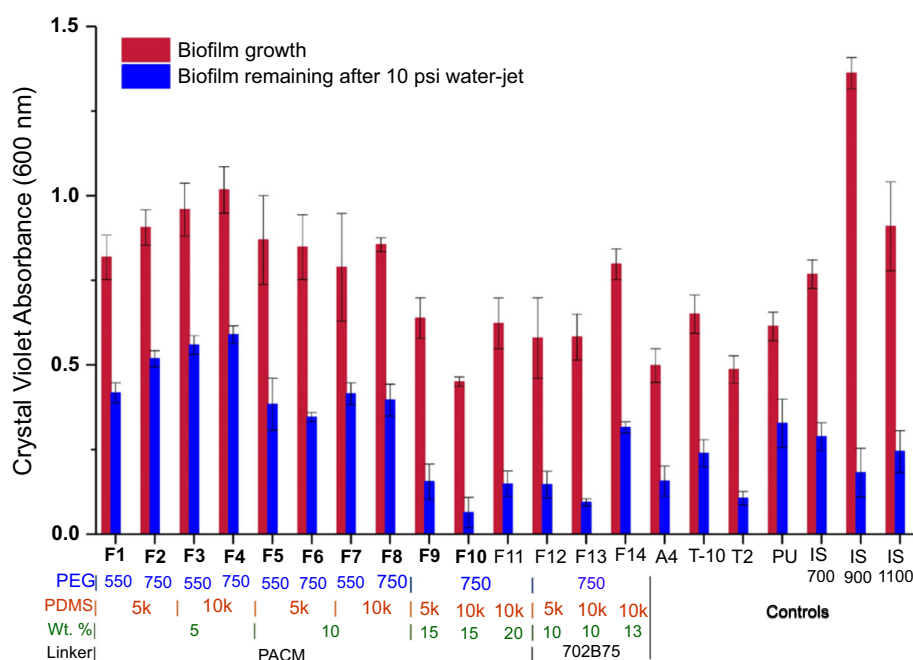


Fig. 8: *C. lytica* data for biofilm growth (red bar) and release at 10 psi (blue bar). The X-axis is labeled to specify the formulation and its components; PEG MW, PDMS MW, wt.% of PEG and PDMS, and curing agent type. The error bars are standard deviations of six measurements

(Fig. 9; comparison P-values < 0.05, Tukey's method). While IS 900 and IS 1100SR coatings showed the least diatom biofouling, coatings F10-F14 showed less or similar biofouling to the T-10 control and IS 700 reference coating. The release of diatom film was evaluated at 10 psi and 20 psi water pressure levels, and a similar trend was observed at both pressures (Fig. 9). The results indicated two differing trends for systems depending on the type of curing agent used. For PACM-cured systems, coatings F1-F3, containing the lowest amount of amphiphilic chains of PEG and PDMS at 5 wt.%, demonstrated a better release than several of the controls, including hydrophobic A4 and amphiphilic T-10 systems, as well as commercial systems IS 700 and PU (comparison P-values < 0.05, Tukey's method). Similar in performance to F1-F3 systems were the F12 and F14 coatings that were cured with 702B75 curing agent and contained amphiphilic chains higher than 10 wt.%. Overall, several AmpSiGC coatings (F1-F3, F12, and F14) demonstrated promising performance, better than both internal control coatings and several commercial systems.

Barnacles are another major macrofouling organism that can cause severe biofouling on surfaces.^{2,67,68} There are many fouling barnacle species (estimated to be 1400 types) and differences in their settlement affinities for surfaces makes it challenging to draw broad conclusions (i.e., preference for hydrophilic vs hydrophobic surfaces).^{61,62,69–74} Due to the limited availability of barnacles, only a few coatings out of the 14 formulations were selected for assessment, but these covered all coating design features. The number of

barnacles that reattached to the AmpSiGC coatings was higher than the control IS 1100SR (probably due to higher density of hydrophilic domains on the AmpSiGC surfaces). However, it was noticed that the barnacles were more easily released from the F4, F6, F8, and F10 AmpSiGC systems, than from the IS 1100SR commercial coating (comparison P-values < 0.05, Tukey's method), suggesting that the amphiphilic systems function effectively as FR surfaces (Fig. 10).^{27,75} A closer analysis indicated that PACM-cured coatings with a concentration of amphiphilic chains between 10 and 15 wt.% of PEG and PDMS offered the best performance. Coatings with amphiphilic concentrations either lower or higher than this range gave a slightly worse performance, but were still better than the commercial reference coatings. The 702B75-cured F14 coating performed worse than PACM-cured F11 coating (both systems contained the highest amount of amphiphilicity in their category), correlating with previous data that showed the curing agent impacts FR output. In summary, systems with PACM curing agent, 10,000 $\overline{M}_n$ PDMS, 750 $\overline{M}_n$ PEG, and amphiphilicity between 10 and 15 wt.% provide a desirable performance against barnacles. Furthermore, comparing the barnacle FR data with ice-release data from the previous study,³⁹ a matching trend was noticed – the optimal FR and icephobic performance was observed for coatings having 10–15 wt.% of PEG and PDMS. Other published studies comparing the performance of an amphiphilic surface for both FR and antiicing applications^{27,40} have also shown the same trend between barnacle and ice release. The results of

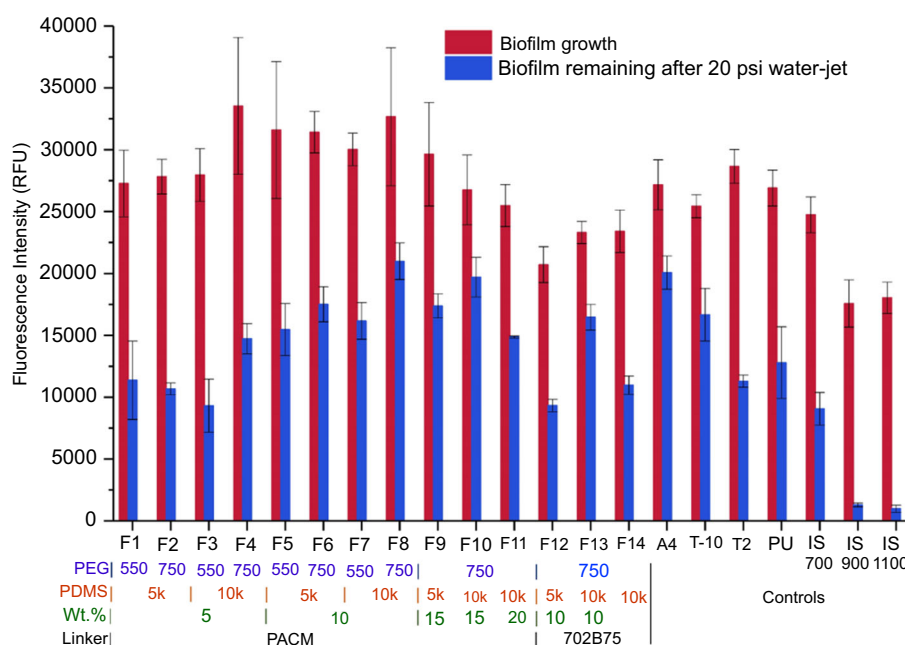


Fig. 9: *N. incerta* data for biofilm growth (red bar) and release at 20 psi (blue bar). The X-axis is labeled to specify the formulation and its components I; PEG MW, PDMS MW, wt.% of PEG and PDMS, and curing agent type. The error bars are standard deviations of six measurements

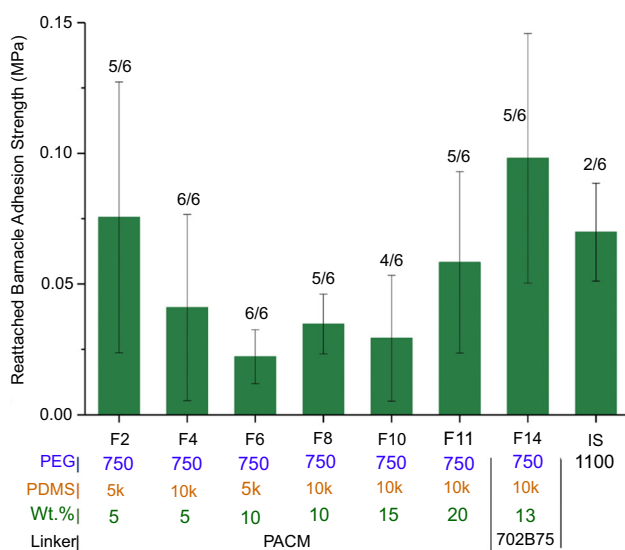


Fig. 10: Reattached barnacle (*A. amphitrite*) adhesion strength data. Six barnacles were used for each reattachment. The number of attached barnacles out of six is labeled as a ratio for each system. Each bar shows the average adhesion strength based on the number of successfully reattached barnacles. The X-axis is labeled to specify the formulation and its components I; PEG MW, PDMS MW, wt.% of PEG and PDMS, and curing agent type. The error bars are standard deviations

this study and others in the literature necessitate further investigation to determine the mechanism and correlation between adhesion of barnacles and ice.

Conclusions

We have reported a new method to develop amphiphilic glycidyl-carbamate-based (urethane-based) coatings by converting the isocyanate groups of the IPDI trimer resin to epoxy functional groups and by introducing PEG and PDMS chains to the IPDI resin. This facile synthesis could potentially eliminate the presence of the isocyanate groups from the final product/formulation, reducing health hazards to workers as well as coating variability due to environmental humidity. Formulations in this study considered four factors of interest, including the molecular weight of PDMS, molecular weight of PEG, amount of PEG and PDMS in a system, and the effect of the curing agent. The surface characterization of the coatings indicated that the objective of creating an amphiphilic surface was achieved. Contact angle measurements showed the presence of a dynamic amphiphilic surface in comparison to a hydrophobic system. ATR-FTIR showed the presence of PEG and PDMS signals on the surface and XPS demonstrated self-stratification of PDMS-based moieties in relation to the amount of amphiphilic prepolymers incorporated in the coating. Additionally, AFM confirmed the presence of heterogeneous domains and that their presence was attributable to the amphiphilic nature of the surface. The heterogeneous patterns were missing from the control hydrophobic system.

In terms of application, the AmpSiGC coatings demonstrated desirable performance as amphiphilic FR coatings. Considering the FR performances, it was generally observed that systems with 15 wt.% or higher

concentration of amphiphilic moieties on the surface performed best (better than internal/external standards) against *U. linza*, *C. lytica*, and barnacles, regardless of the crosslinking agent used. For *N. incerta* (diatom), the FR performance was dependent on the type of curing agent. It can be summarized that the design of amphiphilic siloxane-GC systems can be tuned for a desirable performance by considering the following factors: 1) a 10,000 $\overline{M}_n$ PDMS is favored over lower molecular weights of PDMS – this accords with previously reported work;^{9,27,76} 2) a 750 $\overline{M}_n$ PEG offers relatively better performance than 550 $\overline{M}_n$ PEG; 3) amounts of hydrophobic and hydrophilic moieties in a system affect the FR performance – the preferred amount is between 10 and 15 wt.% or higher depending on type of curing agent; 4) the type of curing agent has a strong impact on FR performance. Furthermore, we demonstrated a correlation between anti-icing³⁹ and FR properties for these amphiphilic systems, which warrants further research.

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